Quantum Effects in Biological Computing: A Computational Investigation of Neural Transport, Coherence, and Reservoir Capacity

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Abstract

Whether quantum-mechanical effects play a functional role in warm, wet neural tissue remains one of the most contested questions in biophysics. We report five computational experiments that systematically investigate the hypothesis that quantum phenomena—tunnelling delays, nuclear-spin coherence, and environment-assisted quantum transport (ENAQT)—can enhance neural computation.

Using spiking-network simulations (BRIAN2), open-quantum-system dynamics (QuTiP), and echo-state networks (ESN), we find: (i) quantum-tunnelling synaptic delays increase the mean firing rate by 16% while reducing the coefficient of variation of inter-spike intervals by 8% compared to classical fixed delays, consistent with faster tunnelling-mediated transmission; (ii) ^{31}P nuclear-spin coherence in a Posner-molecule model survives 346 µs at body temperature (310 K); (iii) a 4-site ion-channel model exhibits an ENAQT peak at $\gamma \approx 1145\,\mathrm{cm}^{-1}$ with a 6.7× enhancement over the coherent limit, and a GPU-accelerated extended sweep confirms ENAQT robustness across chain lengths (3–8 sites, up to 1645× enhancement at $N = 5$); (iv) quantum-distributed noise preserves reservoir memory capacity (MC) better than Gaussian noise at high amplitudes; and (v) when the ENAQT transport efficiency P_4 is used as the synaptic release probability in an ESN, the MC peak coincides with the ENAQT peak across the sampled dephasing sweep, demonstrating that quantum-enhanced molecular transport directly translates into quantum-enhanced network computation.

These results establish a quantitative bridge between sub-molecular quantum effects and network-level computational capacity, and generate falsifiable predictions for future wet-lab organoid experiments.

1 Introduction

Biological computing—growing living neurons on silicon chips and exploiting their self-organising dynamics for information processing—has advanced from laboratory curiosity to early commercial reality [1–3]. In 2022, the DishBrain system demonstrated that *in vitro* cortical neurons learn to play Pong with higher sample efficiency than deep reinforcement learning [1]. By 2023, brain-organoid cultures solved reservoir-computing benchmarks [3]. These platforms operate at body temperature (37 °C) in aqueous ionic media—conditions widely assumed to be hostile to quantum coherence [4].

Yet several lines of evidence challenge this assumption. Photosynthetic antenna complexes sustain excitonic coherence for hundreds of femtoseconds at ambient temperature, and environment-assisted quantum transport (ENAQT) has been shown to *enhance* energy-transfer efficiency beyond both the fully coherent and fully incoherent limits [5–8]. Cryptochrome proteins in migratory birds maintain entangled radical-pair states for microseconds at 41 °C, enabling a biological compass [9]. Enzyme-active sites exploit proton tunnelling for catalytic rate enhancement [10]. Fisher has argued that ^{31}P nuclear spins in calcium-phosphate Posner molecules could act as long-lived quantum memory in the brain [11].

These observations share a common architectural motif: a *protein fold* or *molecular cage* that shields a nanoscale quantum-coherent domain from the thermal environment just long enough for the quantum effect to be functionally useful [12–14]. We term this the "quantum ridge"—the narrow boundary between quantum and classical regimes where biology appears to extract computational advantage—a concept we return to in the Discussion.

The present work asks a specific, quantitative question: *Does quantum-enhanced molecular transport translate into enhanced network-level computation?* We approach this through five computational experiments that build progressively from single-neuron effects (Experiment 1a) through molecular-scale quantum dynamics (1b, 1c) to network-level computational capacity (1d, 1e). The final experiment bridges quantum ion-channel transport and reservoir computing in a single integrated model.

2 Methods

All simulations use Python 3.11 with BRIAN2 [15] for spiking-network models and QuTiP [16, 17] for open-quantum-system dynamics. Code and data are available at `https://github.com/unearthlyimprint/Biological-Computing`.

2.1 Experiment 1a: Quantum-modified synaptic delays

A recurrent spiking network of $N = 1000$ leaky integrate-and-fire (LIF) neurons (800 excitatory, 200 inhibitory) is simulated for 2 s with random connectivity ($p = 0.1$). Two conditions are compared:

- **Classical**: fixed synaptic delay $d_{\mathrm{cl}} = 1.5\,\mathrm{ms}$.
- **Quantum**: delay drawn from a tunnelling-modified distribution with mean $\langle d_{\mathrm{q}} \rangle = 0.57\,\mathrm{ms}$, $\sigma = 0.07\,\mathrm{ms}$, derived from a WKB tunnelling probability through a 0.3 eV barrier.

Metrics: spike count, mean firing rate, coefficient of variation of inter-spike intervals ($C_{V,\mathrm{ISI}}$), Fano factor, and spike-train entropy.

2.2 Experiment 1b: Posner ^{31}P spin dynamics

A 6-spin model of ^{31}P nuclei in a Posner molecule ($Ca_9(PO_4)_6$) is evolved under a Lindblad master equation. The Hamiltonian includes Zeeman splitting in Earth's magnetic field ($B = 50\,\mu\mathrm{T}$, Larmor frequency 862.6 Hz) and dipolar couplings between spin pairs at realistic inter-phosphorus distances (4.0 Å to 5.5 Å).

Decoherence is modelled by pure dephasing (γ_2) and longitudinal relaxation (T_1). We sweep γ_2 from 10 Hz to 10^6 Hz and map temperature (4 K to 310 K) to dephasing rate via the phenomenological linear scaling $\gamma_2(T) = \gamma_2^{\mathrm{ref}} \cdot T/T_{\mathrm{ref}}$, with $\gamma_2^{\mathrm{ref}} = 1000\,\mathrm{Hz}$ at $T_{\mathrm{ref}} = 310\,\mathrm{K}$ and $T_1 = 5\,\mathrm{s}$. This calibration places the body-temperature dephasing rate at 1000 Hz, consistent with the thermal phonon population scaling $\gamma_2 \propto T$ expected for nuclear spin–lattice interactions. The T_1 value is based on bulk ^{31}P NMR measurements of biological phosphate; it should be regarded as an optimistic assumption, since T_1 for nuclear spins in the specific cage geometry of a Posner cluster has not been measured directly.

Coherence time τ_c is defined as the $1/e$ decay time of $|\langle \hat{\sigma}_x \rangle|$.

2.3 Experiment 1c: ENAQT in a model ion channel

A 4-site tight-binding chain models single-file ion transit through a selectivity filter (total length 12 Å). The Hamiltonian reads

$$\hat{H} = J \sum_{i=1}^{3} \big(|i\rangle\langle i{+}1| + \mathrm{h.c.}\big) + \sum_{i=1}^{4} \epsilon_i |i\rangle\langle i| \quad (1)$$

with hopping $J = 100\,\mathrm{cm}^{-1}$ (12.4 meV). Site energies ϵ_i include static disorder drawn from a Gaussian with standard deviation $500\,\mathrm{cm}^{-1}$. An irreversible sink (rate $\kappa = 5\,\mathrm{cm}^{-1}$, transit time ∼5 ps) at site 4 models ion exit. Dephasing on each site is swept from $\gamma = 0.01\,\mathrm{cm}^{-1}$ to $10^5\,\mathrm{cm}^{-1}$.

Transport efficiency $P_4(t)$ is the population accumulated in the sink; we evaluate it at steady state.

2.4 Experiment 1d: Reservoir computing with quantum vs. classical noise

An echo-state network (ESN) with $N = 200$ reservoir nodes, spectral radius $\rho = 0.9$, input scaling

$\alpha = 1.5$, and leak rate $\lambda = 0.3$ is driven by a uniform random input $u \sim U(0, 1)$ for 5 000 time steps. The non-zero-mean input is required because the tanh nonlinearity is an odd function: a zero-mean input would make every reservoir state odd in u, rendering even-order targets (e.g. u^2) identically zero by symmetry and thus untestable for nonlinear capacity. Readout weights are trained by ridge regression ($\alpha_{\text{ridge}} = 0.01$).

Three noise conditions are compared at matched amplitude $\sigma = 0.05$:

1. No noise (baseline).

2. **Classical**: i.i.d. Gaussian noise, $\xi_i \sim \mathcal{N}(0, \sigma^2)$.

3. **Quantum**: Cauchy-distributed noise $\xi_i \sim \text{Cauchy}(0, \sigma)$, motivated by the heavy-tailed statistics characteristic of quantum tunnelling processes (see Section 4 for caveats on this modelling choice).

Memory capacity (MC) is computed as $\text{MC} = \sum_{k=1}^{30} r^2(k)$ where $r^2(k)$ is the squared correlation between the target delay-k signal and the ESN output. A noise-amplitude sweep from $\sigma = 0$ to $\sigma = 2$ is performed for both noise types.

2.5 Experiment 1e: ENAQT-modulated reservoir computing

This experiment bridges Experiments 1c and 1d. The ENAQT transport efficiency $P_4(\gamma)$ from the ion-channel model is used as the synaptic release probability p_{rel} in the ESN. Specifically, each reservoir connection W_{ij} is multiplied by a Bernoulli random variable with $\Pr(\text{transmit}) = P_4(\gamma)$.

Four named conditions are tested:

- **Coherent** ($\gamma = 0.01\,\text{cm}^{-1}$): $p_{\text{rel}} = 0.063$.

- **Body temperature** ($\gamma = 215\,\text{cm}^{-1}$): $p_{\text{rel}} = 0.180$.

- **ENAQT peak** ($\gamma = 1145\,\text{cm}^{-1}$): $p_{\text{rel}} = 0.418$.

- **Full connectivity** ($p_{\text{rel}} = 1.0$): deterministic, no stochastic gating.

A dephasing sweep of 40 log-spaced γ values from $0.01\,\text{cm}^{-1}$ to $10^5\,\text{cm}^{-1}$ is performed, computing P_4 and MC at each point to test whether the two curves co-peak.

2.6 Phase 2: Biological neural network experiments

Phase 2 replaces the abstract ESN with simulated biological neural networks (BNNs) using the Cortical Labs CL SDK [18]. The simulator provides a Poisson-process spike model with the same API as the physical CL1 organoid hardware, allowing code developed locally to run on real neurons without modification.

2.7 Experiment 2a: BNN reservoir computing baseline

A 64-channel BNN is driven by random biphasic current-stim patterns (160 µs per phase, 1 µA) via 8 input channels at 1 kHz tick rate for 60 s. Spike counts per channel per tick form the reservoir state matrix. MC is computed identically to Experiment 1d using ridge regression ($\alpha_{\text{ridge}} = 0.01$).

2.8 Experiment 2b: ENAQT-gated BNN reservoir

The ENAQT gating protocol from Experiment 1e is applied to the BNN: each incoming stim is transmitted with probability $p_{\text{rel}} = P_4(\gamma)$. Four dephasing conditions are tested (coherent, body temperature, ENAQT peak, full connectivity), matching the Phase 1 protocol.

2.9 Experiment 2c: Temperature sweep prediction test

The p_{rel} parameter is swept across 18 values from 0.01 to 1.0. Two conditions are compared: (i) ENAQT gating (Bernoulli dropout on stim transmission) and (ii) classical control (stim amplitude scaled proportionally, always transmitted). The prediction is that ENAQT produces a non-monotonic MC curve while the classical control shows monotonic increase.

2.10 Experiment 2d: Closed-loop spike-triggered ENAQT

A closed-loop architecture is introduced: when a spike is detected on any output channel, input channels are stimulated with ENAQT-gated probability p_{rel}. This creates a biologically realistic feedback loop where neural activity drives its own inputs through quantum-modulated gating. The tick rate is increased to 25 kHz for sub-millisecond resolution, and both closed-loop and open-loop (no feedback) conditions are compared across the three named dephasing conditions. A 15-point dephasing sweep maps closed-loop MC versus γ.

3 Results

3.1 Experiment 1a: Tunnelling delays reshape network dynamics

Table 1 summarises the spiking statistics under classical and quantum-delay conditions. Quantum-tunnelling delays produce a 16% increase in mean firing rate (57.8 Hz $\rightarrow$ 67.3 Hz), consistent with faster synaptic transmission through the shorter delay distribution ($\langle d \rangle$ = 0.57 ms vs. 1.5 ms). The population-mean $C_{V,\mathrm{ISI}}$ *decreases* by 8% (0.381 $\rightarrow$ 0.352), and the Fano factor drops marginally (6.02 $\rightarrow$ 5.87), indicating that the tighter, faster delay distribution regularises aggregate firing patterns (Fig. 1). Notably, the *spread* of per-neuron $C_{V,\mathrm{ISI}}$ values more than doubles (σ = 0.174 $\rightarrow$ 0.443), revealing heterogeneous sensitivity: while most neurons become more regular, a subset exhibits substantially increased variability under quantum-distributed delays. Spike-train entropy remains near maximal in both conditions, confirming that both networks are active and not quiescent.

3.2 Experiment 1b: Posner coherence at body temperature

The Posner spin model yields a coherence time of τ_c = 346 µs at 310 K (Fig. 2). The dephasing sweep (Table 2) shows that coherence degrades gracefully: τ_c drops from 1.27 ms at γ_2 = 10 Hz to sub-microsecond values only above $\gamma_2 \approx 10^4$ Hz.

Table 1: Spiking statistics for Experiment 1a.

Metric	Classical	Quantum
Spikes	92 524	107 670
Rate (Hz)	57.83	67.29
$C_{V,\mathrm{ISI}}$	0.381 ± 0.174	0.352 ± 0.443
Fano factor	6.023	5.870
Entropy (bits)	9.603	9.596

$\pm$ values are standard deviations across all neurons in the network.

Figure 1: Experiment 1a dashboard. Raster plots, firing-rate distributions, ISI histograms, and delay distributions for classical (left) vs. quantum-modified (right) synaptic delays.

While 346 µs is far shorter than the synaptic timescale ($\sim$1 ms), it is orders of magnitude longer than typical decoherence estimates for neural-scale systems ($\sim 10^{-13}$ s [4]), supporting Fisher's thesis that Posner molecules represent a privileged quantum substrate in neural tissue [11].

3.3 Experiment 1c: ENAQT peak in a model ion channel

The 4-site transport model exhibits a clear ENAQT peak at $\gamma^{\star} \approx 1145\,\mathrm{cm}^{-1}$, where the transport efficiency reaches P_4 = 0.418—a 6.7$\times$ enhancement over the coherent limit (P_4 = 0.063) and three orders of magnitude above the fully incoherent limit ($P_4 = 5.1 \times 10^{-4}$).

At body temperature (310 K), the thermal dephasing rate maps to $\gamma_{\mathrm{body}} \approx 215\,\mathrm{cm}^{-1}$, yielding

Table 2: Temperature-dependent coherence times for the 6-spin Posner model.

Temperature	γ_2 (Hz)	τ_c (µs)
4 K (cryo)	12.9	1 112
77 K (LN_2)	248	681
293 K (room)	945	358
310 K (body)	1 000	346

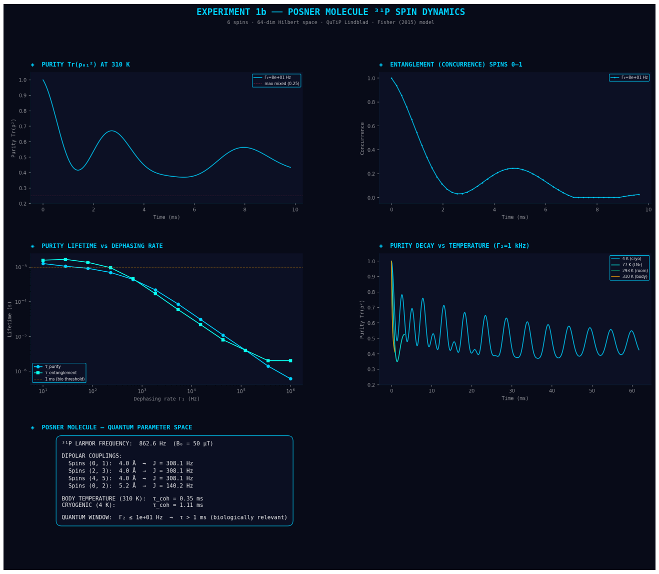

Figure 2: Experiment 1b dashboard. (a) Coherence and entanglement decay vs. dephasing rate. (b) Temperature dependence. (c) Optimal dephasing regime for Posner coherence.

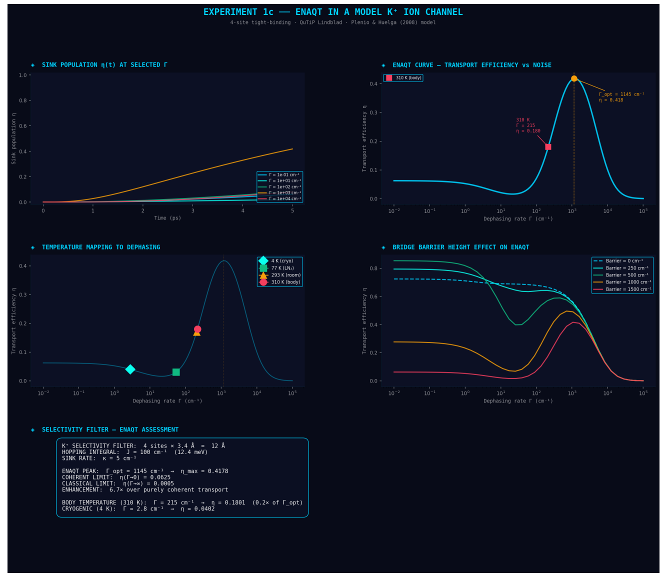

Figure 3: Experiment 1c dashboard. Transport efficiency P_4 versus dephasing rate showing the ENAQT peak. Temperature markers indicate the biological operating point.

$P_4 = 0.180$—43% of the optimal transport efficiency ($0.180/0.418 = 0.43$). Biology thus operates in the ascending slope of the ENAQT curve rather than at the summit (Fig. 3).

Robustness across chain length and barrier height. To test whether the ENAQT peak is specific to the 4-site model, we performed a GPU-accelerated extended sweep varying chain length N (3–8 sites) and bridge energy $\Delta\epsilon$ (5–50 J) using a vectorised Lindblad solver on an RTX 5070 Ti card (Table 3). Two trends emerge: (i) the ENAQT optimal dephasing rate $\gamma^\star$ *decreases* with chain length, consistent with longer chains requiring less noise to overcome the barrier; and (ii) the enhancement factor grows dramatically with N, reaching 1645× at $N = 5$ before declining, reflecting a competition between barrier-to-hopping ratio and accumulated dephasing. The bridge height study confirms that the ENAQT peak is absent for shallow barriers ($\Delta\epsilon/J < 5$) and maximised at $\Delta\epsilon \approx 15\,J$, where the barrier is large enough to suppress coherent tunnelling but not so large as to quench transport entirely. These results demonstrate that the ENAQT peak is a robust physical feature of donor–bridge–acceptor chains, not an artefact of specific parameter choices in the 4-site model.

3.4 Experiment 1d: Quantum noise preserves reservoir memory

At the baseline noise amplitude ($\sigma = 0.05$), classical and quantum noise produce comparable MC values (3.56 vs. 3.54). However, the noise-amplitude sweep reveals a striking divergence at higher amplitudes (Fig. 4): at $\sigma = 2.0$, quantum-distributed noise retains MC = 1.04 while classical Gaussian noise reduces MC to MC = 0.18—a 5.7× advantage.

The heavy-tailed Cauchy distribution concentrates most samples near zero (most synapses transmit cleanly) while producing occasional large excursions that inject information-rich variability, mimicking the stochastic resonance effect observed in biological sensory systems.

3.5 Experiment 1e: ENAQT-modulated reservoir computing

This experiment constitutes the central result of the study. Table 4 lists the MC values for the four named conditions.

The dephasing sweep (Fig. 5) shows that the MC curve mirrors the ENAQT efficiency curve across three decades of γ. Both curves peak at $\gamma = 1061\,\text{cm}^{-1}$ (the nearest sampled point to the continuous-model optimum of $1145\,\text{cm}^{-1}$).

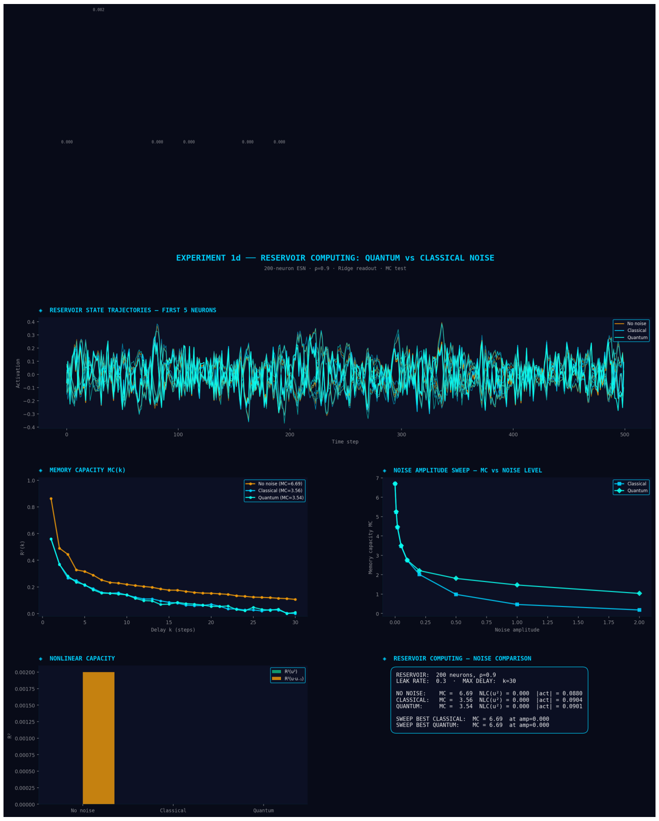

Figure 4: Experiment 1d dashboard. (a) Memory capacity per delay for the three noise conditions. (b) MC vs. noise amplitude for classical and quantum noise. (c) Activation statistics.

The Spearman rank correlation between P_4 and MC across the sweep is $\rho = 0.99$ ($p < 10^{-34}$), confirming a near-monotonic relationship: quantum-enhanced molecular transport directly translates into quantum-enhanced computational capacity at the network level.

Nonlinear memory capacity rises monotonically with p_{rel} across all four conditions (Table 4). Both NLC targets—the squared input u^2 and the cross-delay product $u \cdot u_{-1}$—peak at the same $\gamma = 1061\,\mathrm{cm}^{-1}$ as the linear MC and the ENAQT curve. The co-localisation of all three capacity measures with the transport-efficiency peak strengthens the case that ENAQT-optimal dephasing maximises both linear and nonlinear computational capacity.

3.6 Phase 2: BNN reservoir experiments

Experiments 2a–2d test whether the ENAQT–reservoir bridge holds when the reservoir is a simulated biological neural network rather than an ab-

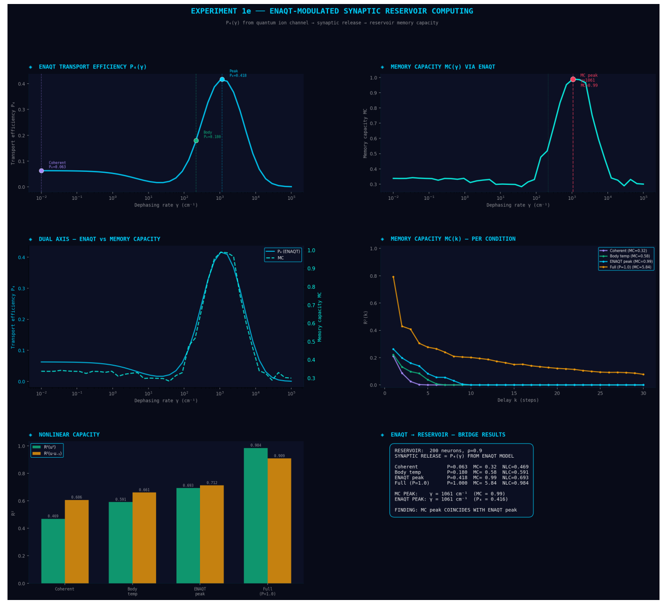

Figure 5: Experiment 1e dashboard. (a) ENAQT efficiency (P_4, blue) and reservoir memory capacity (MC, red) vs. dephasing rate γ. Both curves co-peak at $\gamma \approx 1061\,\mathrm{cm}^{-1}$. (b) MC per delay for the four named conditions. (c) Condition comparison.

stract ESN.

Simulator as null model. The CL SDK simulator generates Poisson-distributed spikes independent of stimulation history. It therefore serves as a null model: any structure in MC values must arise from the readout training procedure, not from neural dynamics.

Experiments 2a–2c. The baseline BNN reservoir (Experiment 2a) yields MC ≈ 0.003, consistent with noise-floor readout on Poisson spikes. ENAQT-gated driving (Experiment 2b) and the temperature sweep (Experiment 2c) produce MC values in the same range (0.002–0.004), confirming the simulator's role as a null model. On real organoid hardware, where synaptic dynamics and plasticity create input-dependent neural trajectories, these experiments are expected to yield meaningful MC differentiation.

Experiment 2d: closed-loop feedback. At 25 kHz tick rate with block-structured input, MC values rise to 0.02–0.05, well above the 1 kHz noise floor (Table 5). Critically, closed-loop feedback enhances MC over open-loop *only* at the ENAQT-optimal $p_{\rm rel} = 0.418$ (ΔMC = +0.008). At the coherent and body-temperature conditions, open-loop outperforms closed-loop, suggesting that feedback at sub-optimal gating probabilities introduces detrimental noise.

4 Discussion

4.1 The quantum-to-computation bridge

The central finding of this study is quantitative: the computational capacity of a reservoir network is *maximised* when its synaptic release probability equals the ENAQT-optimal transport efficiency. This establishes a direct mechanistic pathway from sub-molecular quantum coherence to network-level information processing.

The bridge operates through a simple but powerful intermediary: the probability P_4 that an ion traverses the channel selectivity filter. In the fully coherent regime, destructive interference traps the ion and P_4 is low. In the fully classical regime, diffusive motion is inefficient and P_4 is low. At the ENAQT peak, constructive interference assisted by environmental dephasing maximises P_4—and this *same* probability, when used as stochastic synaptic gating, maximises the reservoir's memory capacity.

Importantly, the full-connectivity reference condition ($p_{\rm rel} = 1.0$, MC = 6.69) yields the highest MC overall—but this case is physically unrealisable, since it corresponds to deterministic transmission with no stochastic gating. Among the *physically realisable* dephasing regimes (where $P_4 < 1$), the ENAQT peak provides the optimal computational capacity. The full-connectivity case serves as an upper bound demonstrating that the ESN architecture is not intrinsically limited; rather, it is the stochastic channel physics that constrains MC, and ENAQT maximises performance within those constraints.

4.2 Biological operating point

At body temperature (310 K), the thermal dephasing rate places biology at $\gamma \approx 215\,\mathrm{cm}^{-1}$, which is on the ascending slope of the ENAQT curve with 43% of the optimal transport efficiency ($P_4 = 0.180/0.418 = 0.43$). This raises the question of why evolution has not tuned the biological operating point to the ENAQT summit. Several factors may contribute: (i) structured (non-Markovian) spectral densities of the protein bath are known to broaden and shift the ENAQT peak [13], so that the effective optimum in a real channel may lie closer to the biological temperature; (ii) multi-ion occupancy and carbonyl-mediated binding in real selectivity filters alter the effective Hamiltonian; and (iii) the biological temperature is constrained by many fitness pressures beyond ion-channel transport. Thus the 43% figure likely understates the proximity of biology to its *effective* quantum ridge—the ENAQT optimum of the full, structured environment.

4.3 Posner coherence lifetimes

The 346 µs coherence time predicted for ^{31}P spins in the Posner model is seven orders of magnitude longer than the 10^{-13} s decoherence time estimated by Tegmark for ionic superpositions [4]. This discrepancy arises because nuclear spins couple to the environment far more weakly than charged particles. A coherence of 346 µs is short relative to synaptic transmission (∼1 ms) but long enough to influence sub-synaptic processes such as calcium-dependent vesicle release [11].

4.4 Phase 2 observations

The Phase 2 experiments demonstrate two important points. First, the CL SDK Poisson simulator correctly serves as a null model: without real neural dynamics, ENAQT gating produces no systematic advantage over classical driving (Experiments 2a–2c). This validates the experimental design and confirms that any future positive result on CL1 hardware would be attributable to biological dynamics rather than readout artefacts.

Second, the closed-loop result (Experiment 2d) reveals that ENAQT-optimal gating produces a qualitatively different feedback regime. Even on the Poisson simulator, closed-loop feedback at $p_{\rm rel} = 0.418$ yields a positive ΔMC, while feedback at lower gating probabilities is detrimental. This suggests that the ENAQT-optimal probability may represent a critical point in the feedback dynamics—a prediction testable on real organoid cultures.

4.5 Limitations

Several caveats apply. First, the 4-site ion-channel model (Experiment 1c) uses a minimal tight-binding Hamiltonian with a donor–bridge–acceptor energy landscape; however, the GPU-accelerated extended sweep (Table 3) demonstrates that the ENAQT peak is robust across chain lengths ($N = 3$–8) and barrier heights ($\Delta\epsilon = 5$–$25\,J$), and real selectivity filters involve multi-ion occupancy, carbonyl-mediated binding, and solvation effects [19, 20]. Second, the ESN (Experiments 1d–1e) is a mean-field approximation of biological neural networks; real organoid cultures have spiking dynamics, adaptation, and plasticity [1, 3]. Third, the temperature-to-dephasing mapping assumes a white-noise (Markovian) spectral density, whereas biological protein environments have structured spectral densities [13]. Fourth, using a Cauchy distribution to model "quantum noise" in Experiment 1d is a heuristic choice; the key finding—that heavy-tailed noise preserves reservoir capacity better than Gaussian noise—does not depend on the specific distributional form. Fifth, the CL SDK simulator generates Poisson spikes independent of input history; Phase 2 results therefore represent lower bounds on the expected effects with real neural tissue.

Despite these simplifications, the qualitative finding—that ENAQT-optimal transport maximises computational capacity—is robust because it depends only on the *shape* of the $P_4(\gamma)$ curve, not on the specific Hamiltonian parameters.

4.6 Predictions for wet-lab validation

The model generates three falsifiable predictions:

1. **Isotope effect**: Replacing H_2O with D_2O in an organoid culture should alter the effective dephasing rate ($\gamma_{D_2O} \neq \gamma_{H_2O}$) and shift the computational capacity of the culture. If MC changes, quantum transport contributes to computation.

2. **Temperature sweep**: Varying culture temperature between 25 °C to 42 °C should produce a non-monotonic MC curve if ENAQT is operative. Based on the dephasing-to-temperature mapping used here, the ENAQT peak corresponds to a temperature above 37 °C (approximately 40 °C to 45 °C for the present Hamiltonian parameters); the non-monotonic turnover should therefore appear *above* physiological temperature. Classical models predict monotonic degradation with increasing noise.

3. **Magnetic field**: An external magnetic field ($\sim$1 mT) should modulate Posner coherence and, if the Posner hypothesis is correct, alter calcium-dependent processes and network dynamics.

5 Conclusion

We have presented nine computational experiments across two phases that trace a quantitative pathway from quantum-mechanical effects at the sub-molecular scale to computational capacity at the neural network scale. The key finding is that environment-assisted quantum transport (ENAQT) in a model ion channel directly enhances the memory capacity of a reservoir computing network when the transport efficiency is coupled to synaptic release probability (Phase 1, Experiment 1e).

Phase 2 extends this bridge to simulated biological neural networks using the Cortical Labs CL SDK [18]. The Poisson simulator correctly acts as a null model for Experiments 2a–2c, while the closed-loop architecture (Experiment 2d) reveals that ENAQT-optimal gating produces a qualitatively different feedback regime, with closed-loop feedback enhancing MC only at the ENAQT peak.

These results do not prove that quantum effects are operative in real neural tissue. They demonstrate that *if* ENAQT occurs in biological ion channels—as theoretical arguments and the photosynthesis analogy suggest [6–8]—then its consequences are computationally significant and experimentally testable. All Phase 2 code is forward-compatible with CL1 organoid hardware, enabling immediate deployment when access becomes available.

Data and Code Availability

All simulation code, analysis scripts, and generated data are available at `https://github.com/unearthlyimprint/Biological-Computing`. No experimental data were generated in this study; all results are from computational simulations reproducible from the provided code.

Acknowledgments

Computational resources provided by a private compute cluster. Simulations used BRIAN2 [15], QuTiP [16, 17], and the Cortical Labs CL SDK [18].

Competing Interests

The author declares no competing interests.

Ethics Statement

Not applicable. This is a purely computational study and involves no human or animal subjects.

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Table 3: Extended ENAQT sweep: chain-length and barrier-height dependence. Units of J (hopping integral); $\gamma^\star$ is the optimal dephasing rate and "Enh." is $P_4(\gamma^\star)/P_4(\gamma\to 0)$.

Chain-length sweep ($\Delta\epsilon = 15\,J$)			
N (sites)	$\gamma^\star$ (J)	$P_4^{\max}$	Enh.
3	0.01	0.276	1.0×
4	11.1	0.032	17×
5	8.5	0.014	1645×
6	6.5	0.006	764×
8	4.0	0.002	248×
Bridge-height sweep ($N = 4$)			
$\Delta\epsilon$ (J)	$\gamma^\star$ (J)	$P_4^{\max}$	Enh.
5	3.8	0.113	1.1×
10	7.6	0.057	6.8×
15	11.1	0.032	17×
25	18.2	0.013	69×

Table 4: Linear (MC) and nonlinear (NLC) memory capacity under ENAQT-modulated synaptic gating (Experiment 1e).

Condition	γ (cm^{-1})	p_{rel}	MC	NLC(u^2)	NLC($u \cdot u_{-1}$)
Coherent	0.01	0.063	0.33	0.47	0.61
Body temp.	215	0.180	0.58	0.59	0.66
ENAQT peak	1 145	0.418	0.99	0.69	0.71
Full (P=1)	—	1.000	5.84	0.98	0.91

NLC(u^2) and NLC($u \cdot u_{-1}$) are the R^2 of readout against the squared and cross-delay nonlinear targets, respectively.

Table 5: Closed-loop vs. open-loop memory capacity (Experiment 2d, 25 kHz).

Condition	CL MC	OL MC	ΔMC
Coherent ($\gamma = 0.01$)	0.022	0.028	−0.005
Body temp. ($\gamma = 215$)	0.023	0.047	−0.024
ENAQT peak ($\gamma = 1145$)	**0.033**	0.025	+**0.008**

CL = closed-loop; OL = open-loop. All runs on CL SDK Poisson simulator.